



AIAA3201

Introduction to Computer Vision

Week 1

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Thrust, INFO Hub



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联系方式

- WANG Hao
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- TA
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务必自带笔记本
电脑!

WeChat 群



群聊：AIAA 3201



该二维码7天内(9月7日前)有效，重新进入将
更新

Today's Journey



Cool Applications



History of CV



Course Plan



Image Basics

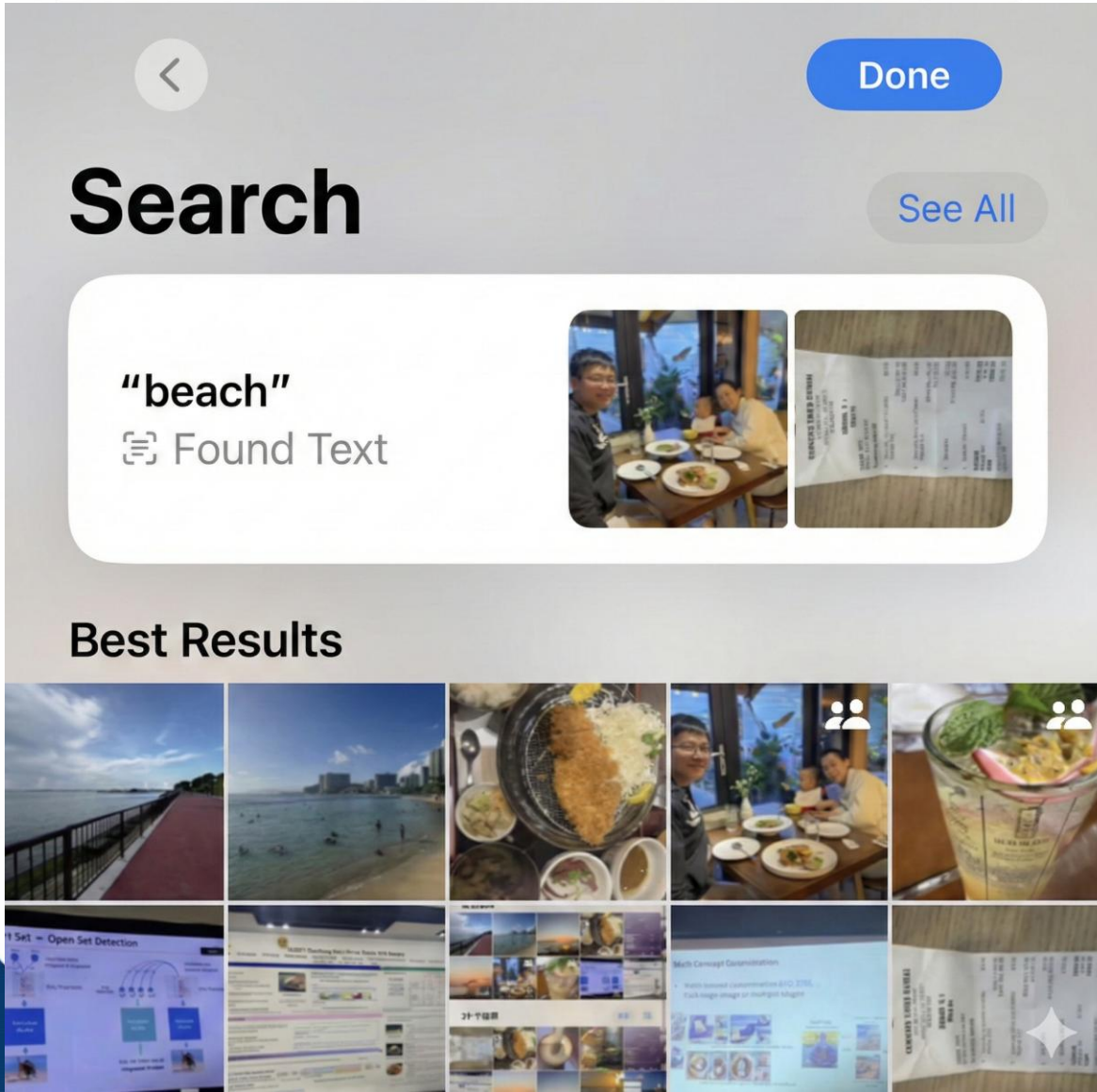
Let's Start



打开 手机中的照片应用并搜索一些内容，
例如，海滩。



Smart Photo Album Search



Search: Beach → Results

No Manual Tags

AI Understands Photos



How AI Sees?

You: 🏖️ Beach



How AI Sees?

You: 🏖️ Beach



Computer: [123, 45, 67...]

[123]	45	67]	[16]	[44]	[45]	67]
67]	[100	20	50	[34]	[49]	50]
[100	50]	[100	20	[37]	[45]	50]
[100	[120	50]	[30]	[32]	45	67]
[200	[150]	25]	[30]	[45]	[30]	67]
[200	[150]	25]	[30]	[45]	[36]	[37]

How AI Sees?

You: 🏖️ Beach



Computer: [123, 45, 67...]

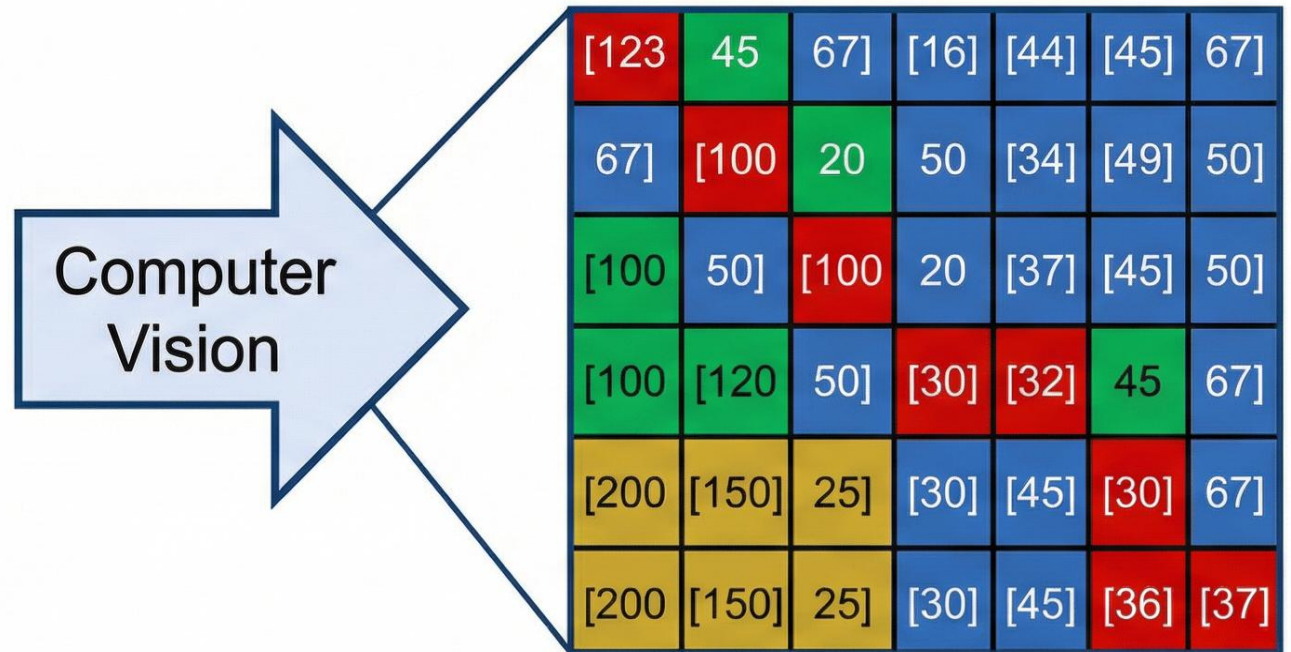


Image → Numbers → AI → Understanding

CV: Numbers → Understanding



Autonomous Driving



- Real-time:
 - 🚗 Vehicles
 - 🚶 Pedestrians
 - 🛣️ Lanes
 - 🚦 Signs

30+ FPS

Source: HyperChange

超越感知，计算机视觉如今能够进行**视觉推理**！





定位证据整合



关键标识符分析

1.“You Yuan Cooked Food”招牌（置信度：0.44）：

- 中英文名称：友源熟食 / YOU YUAN COOKED FOOD • 新加坡小贩中心摊位的典型命名风格

2.“Fine Met Cooked Food”招牌（置信度：0.41）：

- 中英文名称：佳美熟食 / Fine Met Cooked Food • 与新加坡多元文化饮食文化的特征相符

3.“鲜榨果汁”摊位（置信度：0.38）：

- 带有价签“100”（可能表示 \$1.00）
- 新加坡小贩中心常见的一种饮料摊位



地理定位的关键证据

1.Architectural Feature Verification:

- 金属天花板结构
- 集中式用餐区布局 • 吊扇通风系统 → 新加坡小贩中心的典型设计

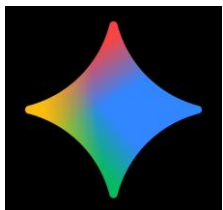
2.文化标识:

- 中英双语标识牌
- 烧腊等中餐 → 新加坡华人社群的特征

3.商业环境:

- 露天用餐空间
- 集中管理的熟食摊位 → 与建屋发展局（HDB）的小贩中心模式一致





最终定位结论

✅ **确认地点：** Challenger Way 小贩中心（Challenger Way熟食中心），位于新加坡裕廊（裕廊区）

✅ **具体特征：**

- 位于 **HDB**（组屋区）一栋楼的底层
- 官方命名的 **小贩中心**（小贩中心）
- 邻近 **Jurong East MRT 站**（邻近裕廊东地铁站）

📌 **验证建议：**

1. 搜索 "**Challenger Way Hawker Centre**" 在 Google Maps 上。
2. 查看街景以匹配 **金属天花板结构** 和 **摊位布局**。
3. 确认诸如 "**You Yuan Cooked Food**" (友源熟食) 和 "**Jia Mei Cooked Food**" (佳美熟食) 等摊位是否存在。

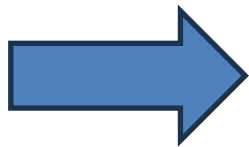


图像/3D/视频 生成





输入：
一只松鼠吃
了一些超辣
面条



猩红守望者 — Bloom of Destruction (猩红守望者 — 毁灭之华) 结构 (可变编辑时长; 总长度 25s; 内容 23.00s + 黑屏 2.00s; 模式: 标准; 节奏因子: 2.0)

[#1 0.05s]

•**ACTION:** 静态镜头 — 一名骑士单膝跪地, 以一把发光的剑支撑着自己, 四周环绕着红花。

•**CAMERA:** 中景镜头, 固定, 居中。

•**主体:** 女骑士; 兜帽下戴着铁面具; 暗淡的金色盔甲; 覆满灰尘的斗篷; 半透明红色光刃; 跪姿。

•**场景:** 破败的地下神庙洞穴; 地面铺满红花; 空气中浮动着尘埃。

•**光线:** 冷光从拱顶的裂隙中倾泻而下; 剑作为红色光源; 空灵的薄雾。

•**CAM:** 中景; 居中对称; 35mm 质感; 固定机位。

[#2 1.00s]

•**ACTION:** 她将剑更深地刺入地面; 嗡鸣声增强; 花瓣颤动。

•**CAMERA:** 从花瓣高度处的低角度推进。—— [动态] /* Qi/Aura */

•**主体:** 覆甲的手紧握剑柄; 铁面具低垂。

•**场景:** 花海中央。

•**光照:** 红色核心光晕; 柔和的边缘光。

...



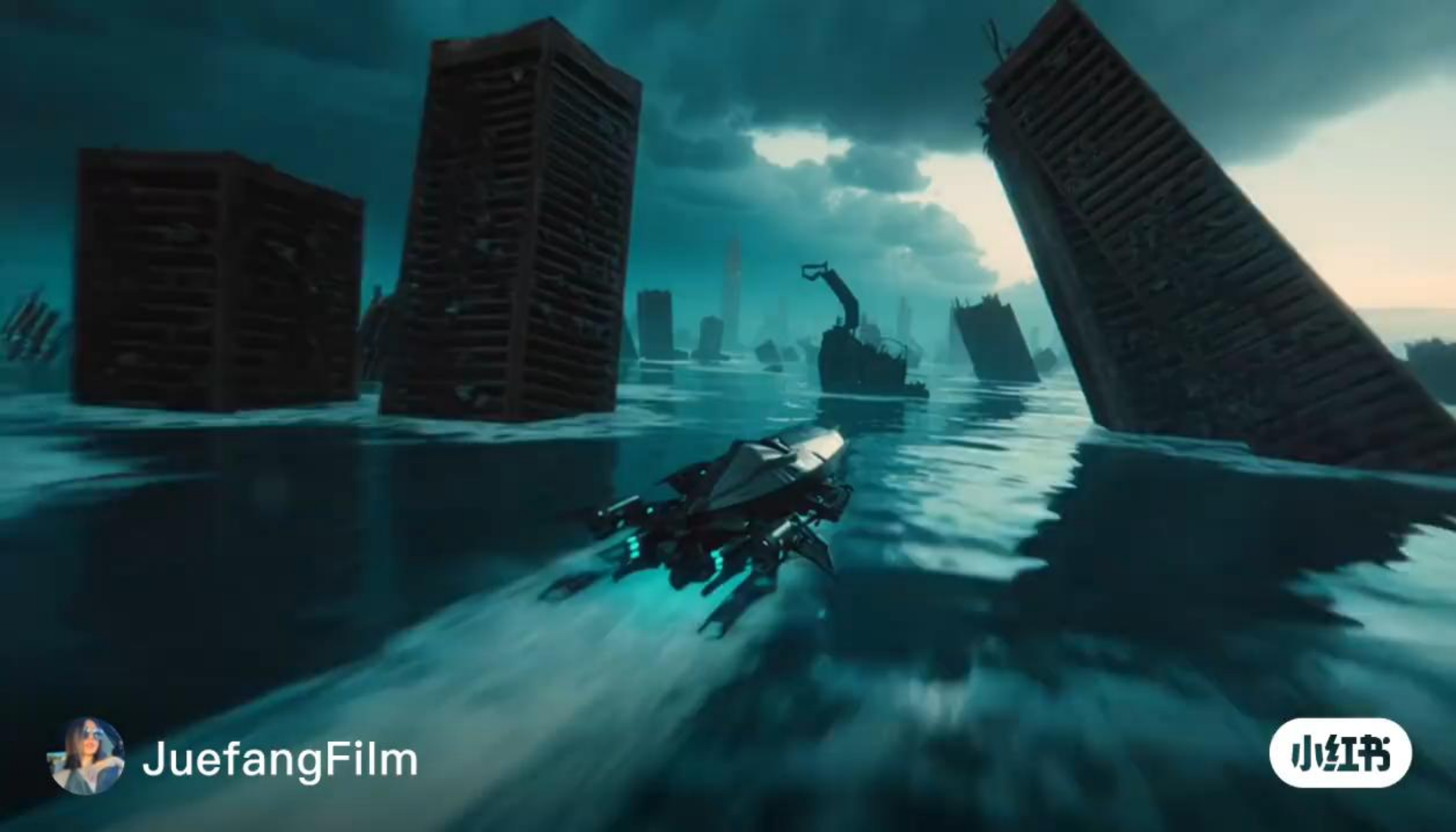
这个领域充满了机遇。

而大多数机会属于年轻人，就像你一样









JuefangFilm

小红书

目前 仍有许多项目可以实现商业化.....

上完这门课后，如有兴趣请联系我。



Part 1

What is Computer Vision?



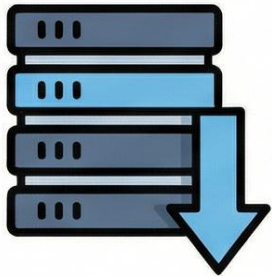
Computer Vision: Definition



Enable computers to **SEE** and **UNDERSTAND** visual world



EXTRACT meaningful information

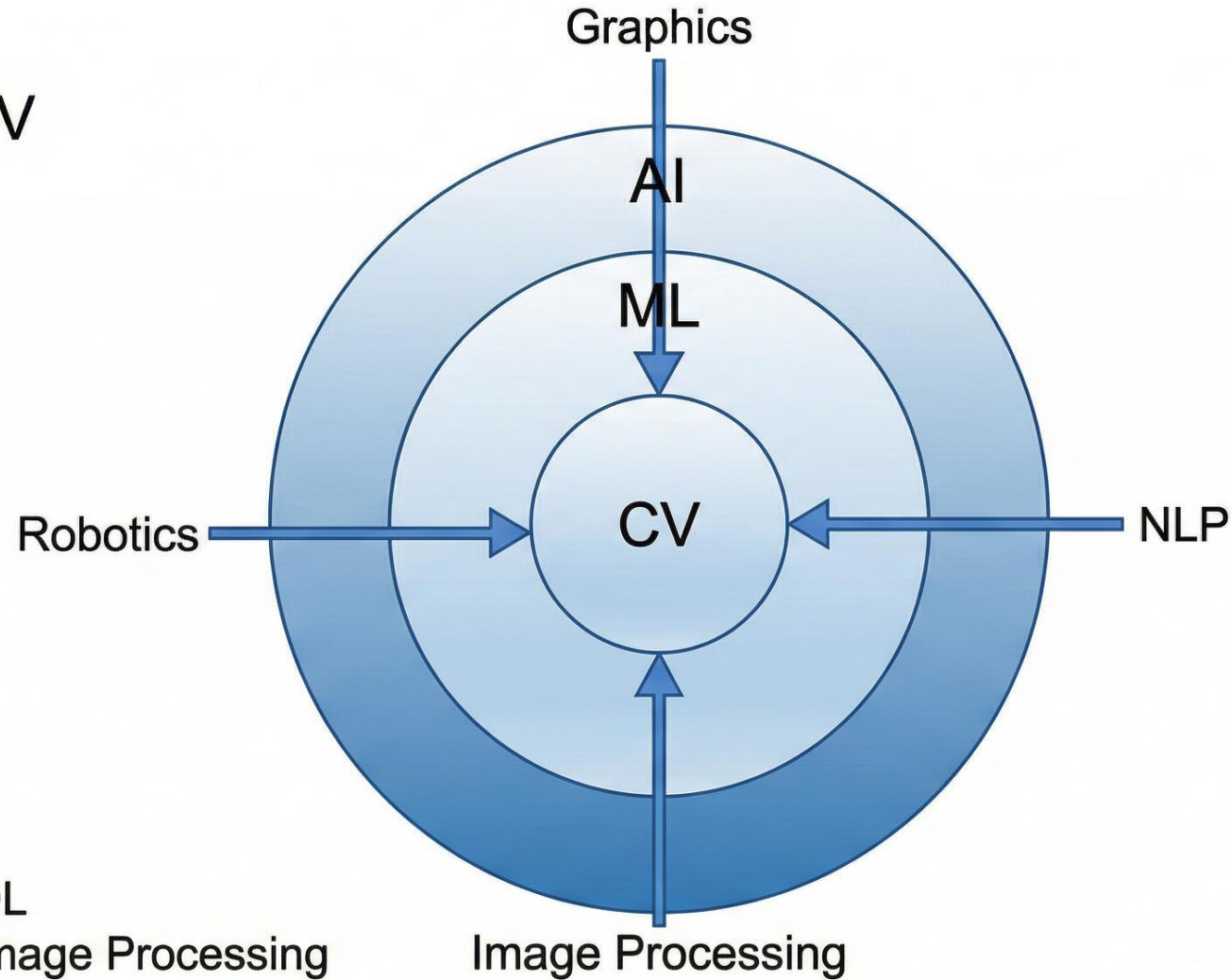


Goal: Human-level **PERCEPTION**



CV in Context

$AI \supset ML \supset CV$



CV uses: ML, DL

CV builds on: Image Processing

CV relates to: NLP, Robotics



Main CV Tasks



Classification:
What?

Main CV Tasks



Classification:
What?



Detection:
Where?

Main CV Tasks



Classification:
What?



Detection:
Where?



Generation:
Create?

Main CV Tasks



Classification:
What?



Detection:
Where?



Generation:
Create?



Segmentation:
Boundaries?

Main CV Tasks



Classification:
What?



Detection:
Where?



Generation:
Create?



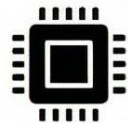
Segmentation:
Boundaries?

+ More: Pose, Depth, Tracking...

<https://huggingface.co/spaces/Ruicheng/MoGe-2>



Quick Recap

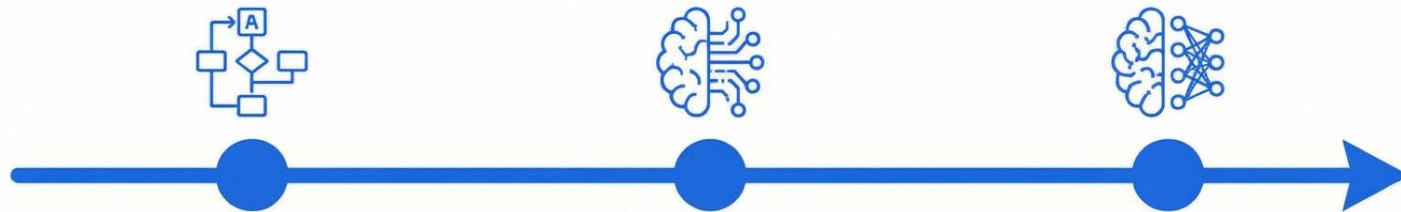
- ✓ CV = Subdomain of AI  
- ✓ Tasks: Understanding  → Generation 
- ✓ Applications: Everywhere   

Next: History →

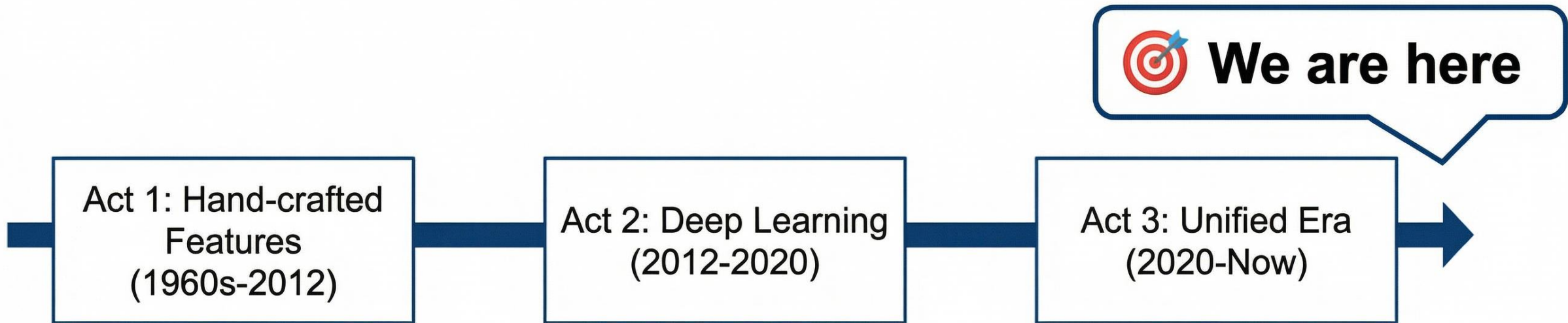


Part 2

History of Computer Vision A Journey Through Three Eras



Three Eras of CV



Act 1

Hand-crafted Features Era

1960s – 2012



MIT Summer Vision Project (1966)

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

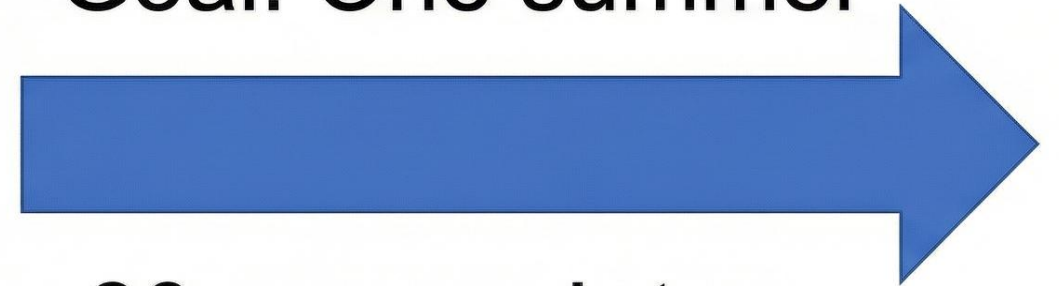
July 7, 1966

THE SUMMER VISION PROJECT

Seymour Papert.

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition!!.

Goal: One summer

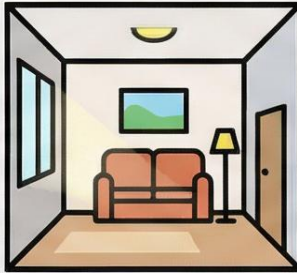
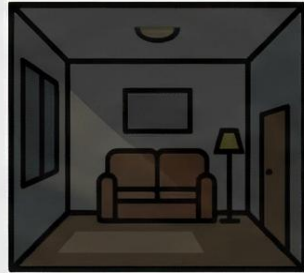


60+ years later...

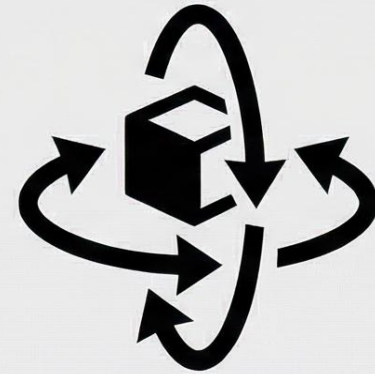
Lesson: Vision is HARD



Why is Vision So Hard?



Lighting



Viewpoint



Occlusion

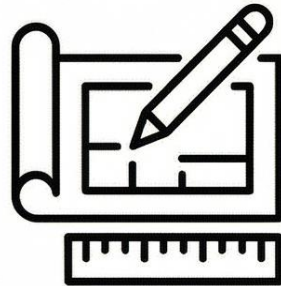
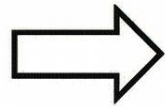


Clutter

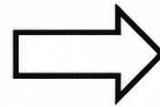
Hand-crafted Approach



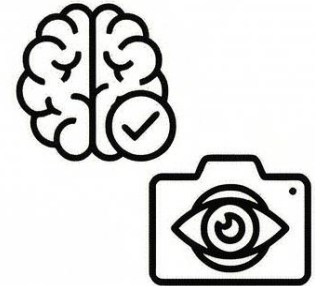
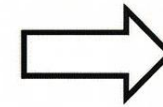
Human Expert



Design Rules

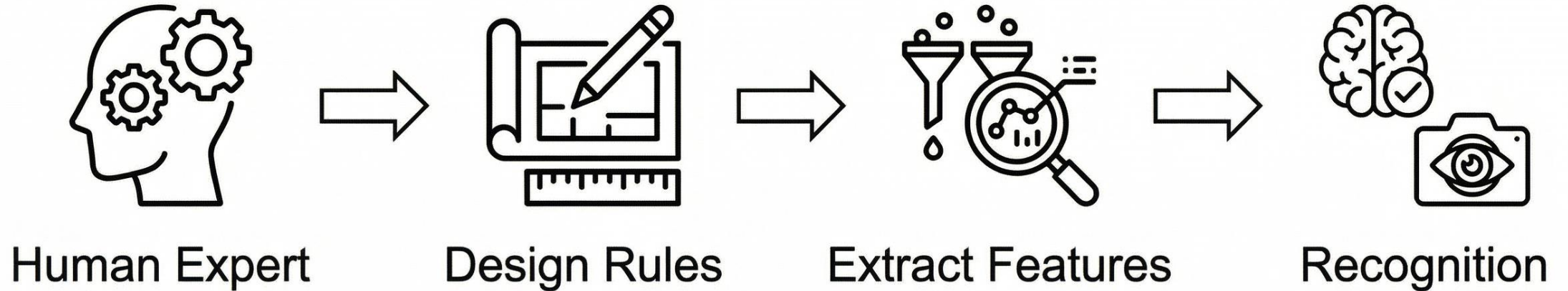


Extract Features



Recognition

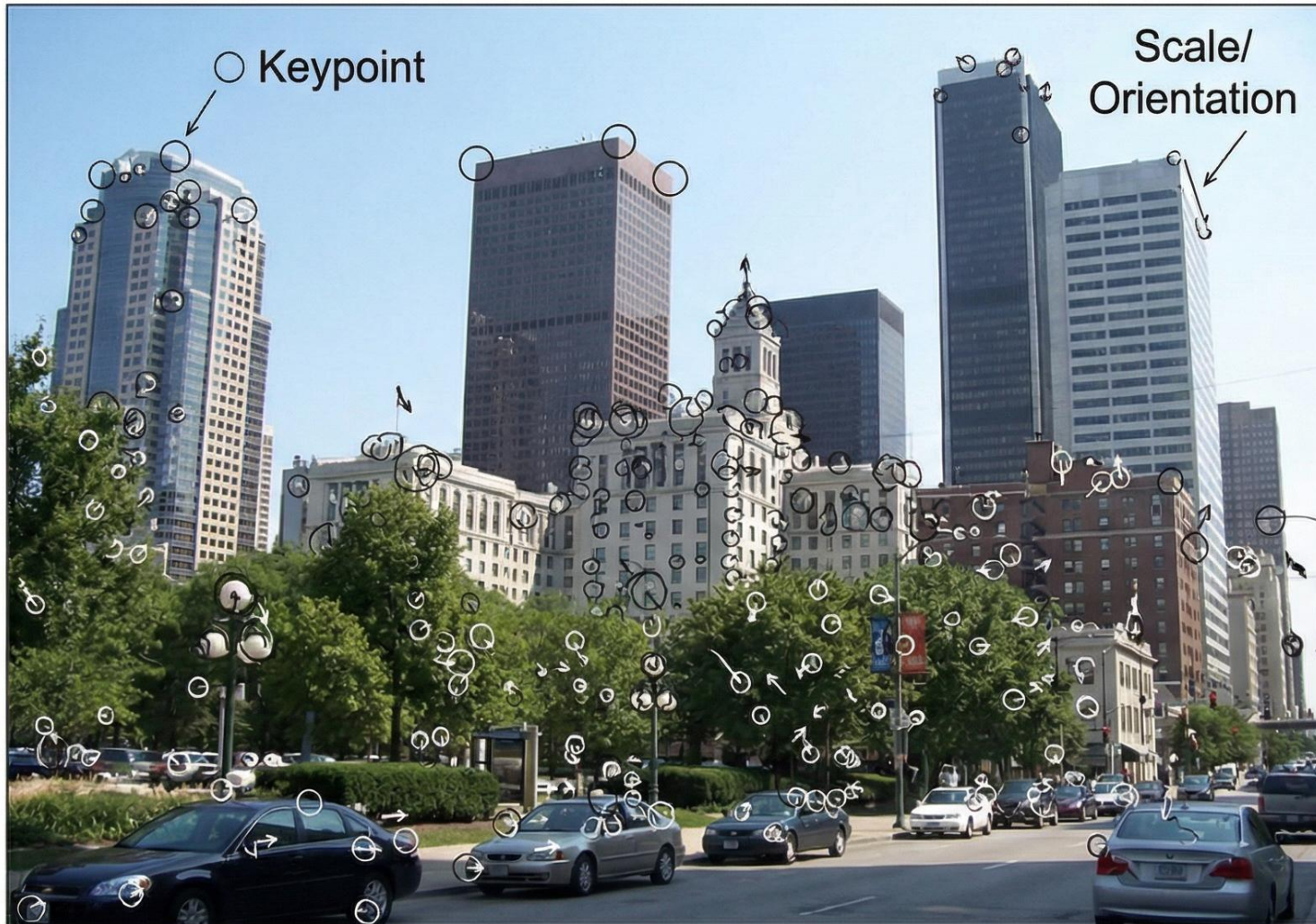
Hand-crafted Approach



Philosophy: Humans tell machines

Examples: SIFT, HOG, Haar

SIFT: Feature Detection



	Scale-invariant
	Rotation-invariant
	Human-designed

SIFT Features

Detects: Keypoints (corners, edges)

Invariant to: Scale, Rotation
Human-designed

Limitations

✗ Task-specific

✗ Poor generalization

✗ Hard to design

✗ Performance ceiling

Breakthrough needed...



Act 1 Summary

1960s-2012
50+ years



Human-designed
rules

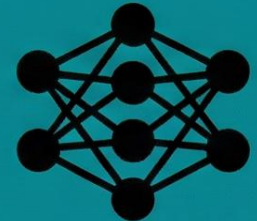


Basic CV
applications



Task-specific
limitations

2012
AlexNet



Then everything changed...

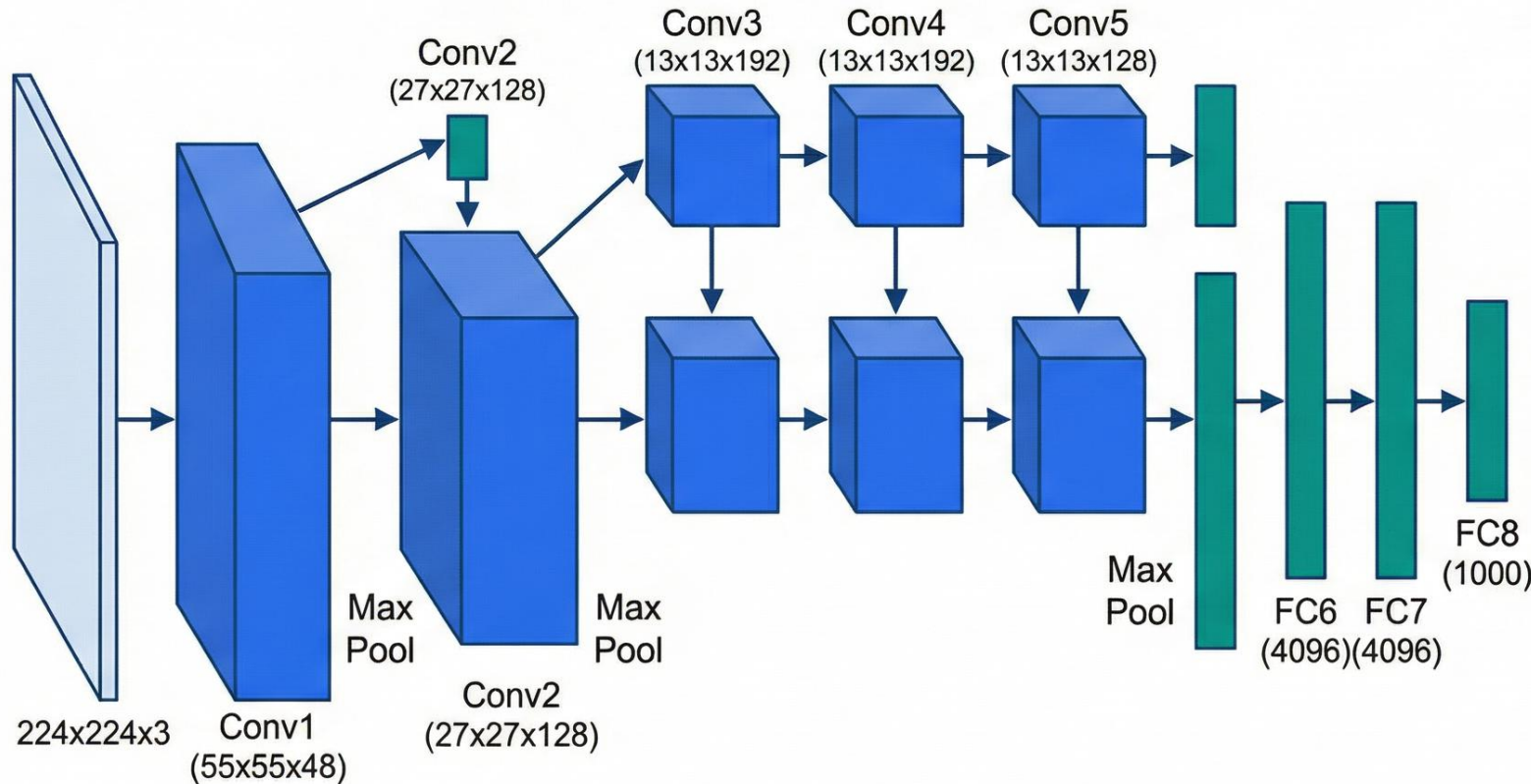
Act 2

Deep Learning Era

2012 – 2020



AlexNet (2012)



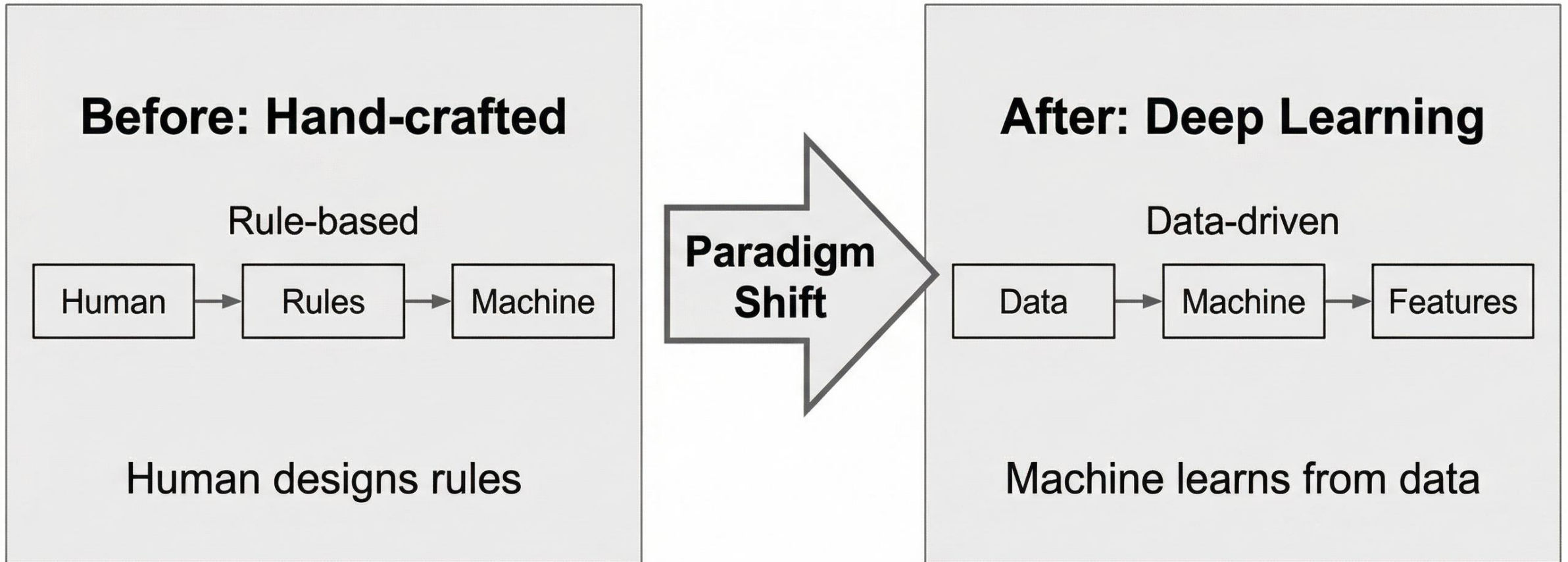
ImageNet Challenge:
AlexNet: 15.3%
2nd Place: 26.2%

Gap: 10%
Everyone Stunned

The Game Changed

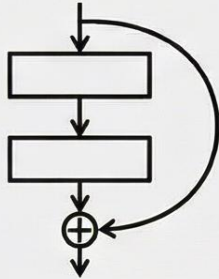


Paradigm Shift



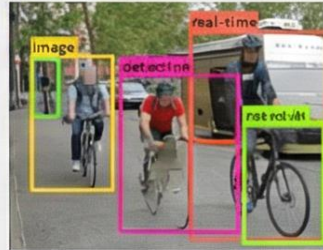
Tell what → Learn how

DL Achievements (2012-2020)



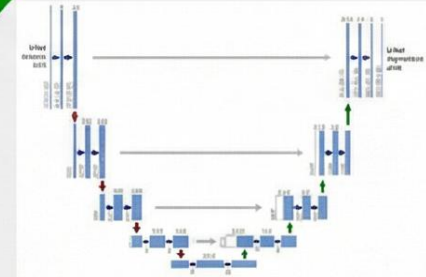
ResNet:

Classification: 95%+



YOLO:

Detection: Real-time



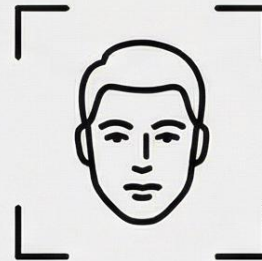
U-Net:

Segmentation: Pixel-level



OpenPose:

Pose: Joints



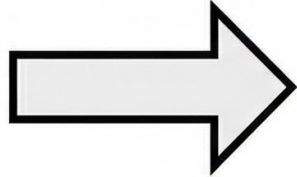
FaceNet:

Face: Beyond human

+ Depth, Video, 3D...

Act 2 Limitations

- ✗ One model, one task
- ✗ Understanding $\neq$ Generation
- ✗ Massive labeled data
- ✗ Lacks generalization

Can we unify them? 



Act 3

Unified Era



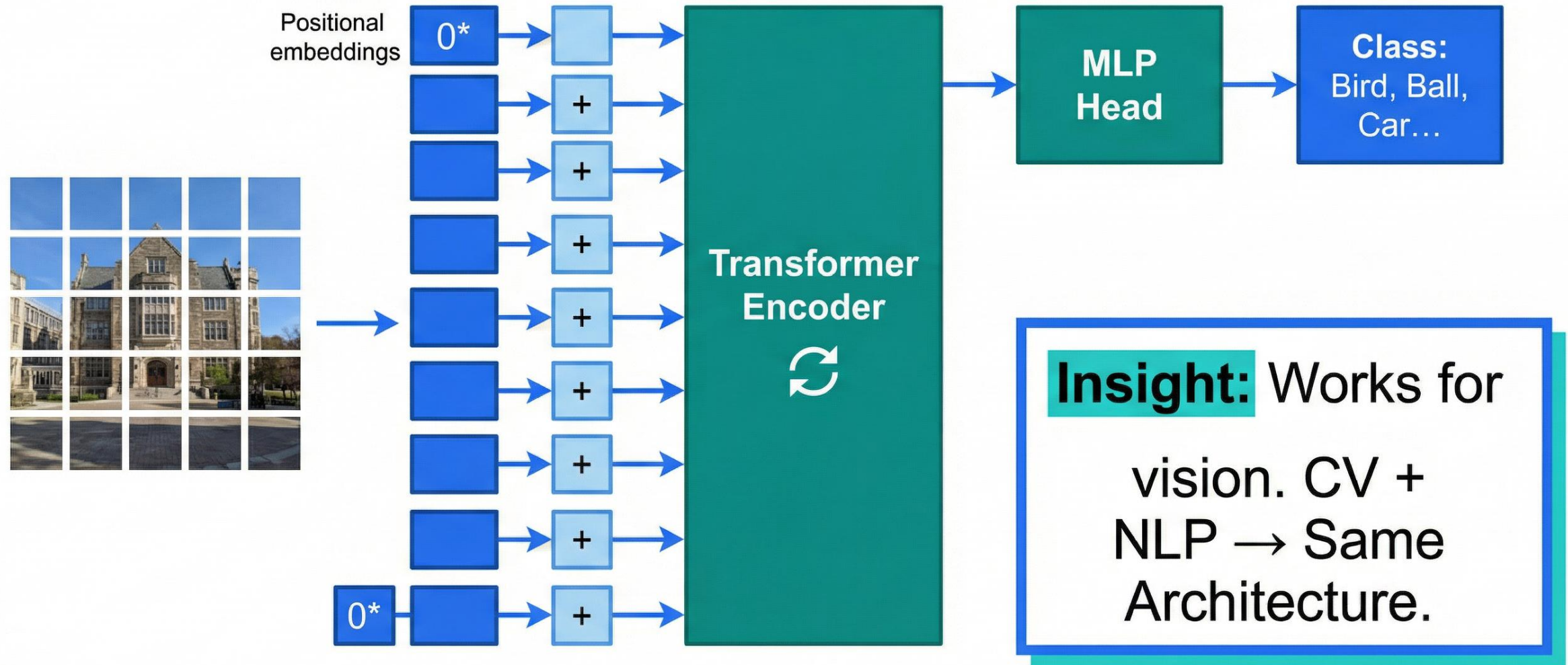
2020 – Now



We are here



Vision Transformer (2020)



Four Unifying Trends

1 Multimodal
Text + Vision

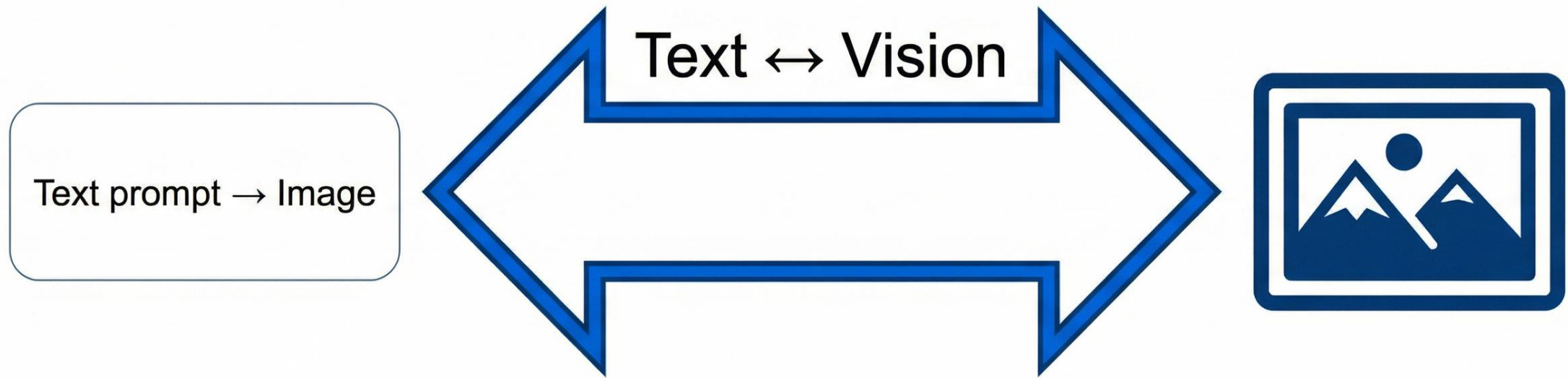
2 Zero-shot
No training

3 Visual Dialog
Reasoning

4 World Model
Unified

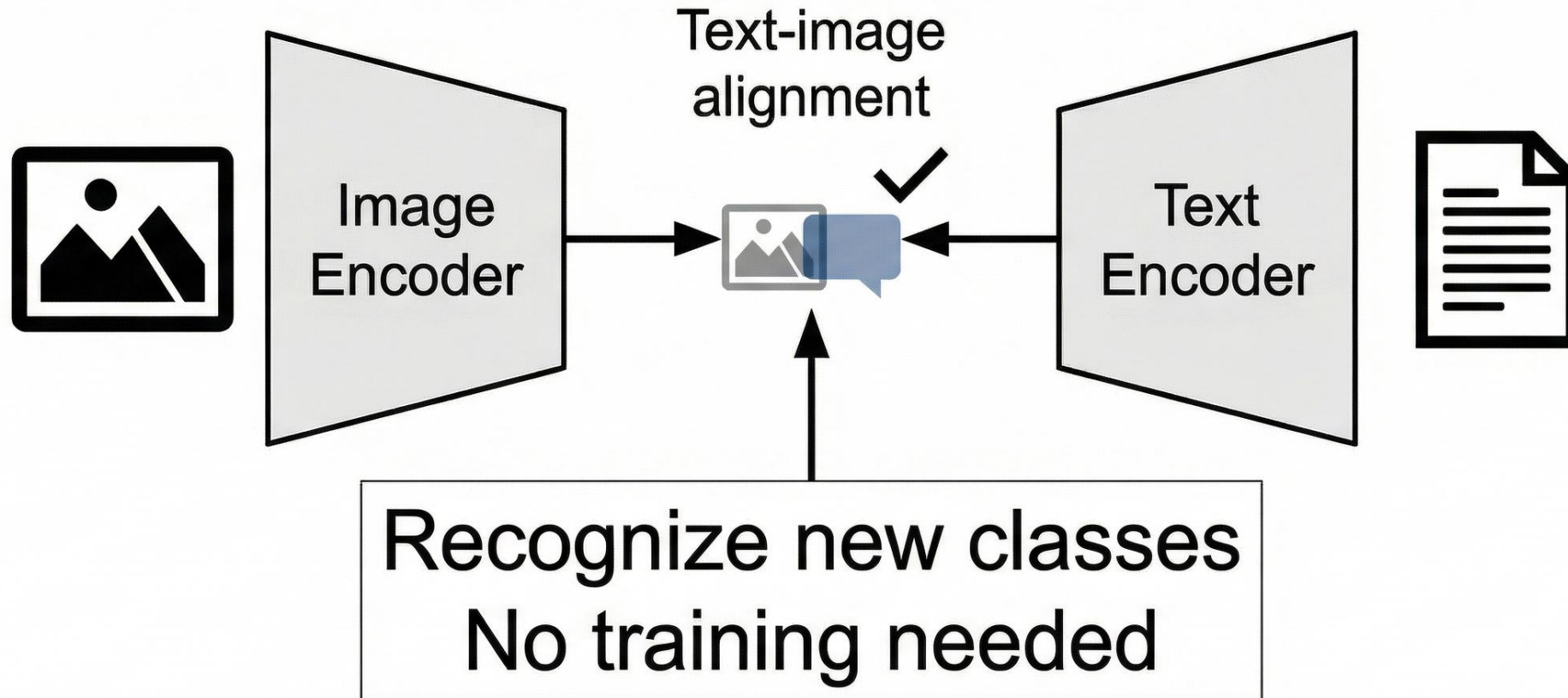


Trend 1: Multimodal



DALL-E, Midjourney, Stable Diffusion

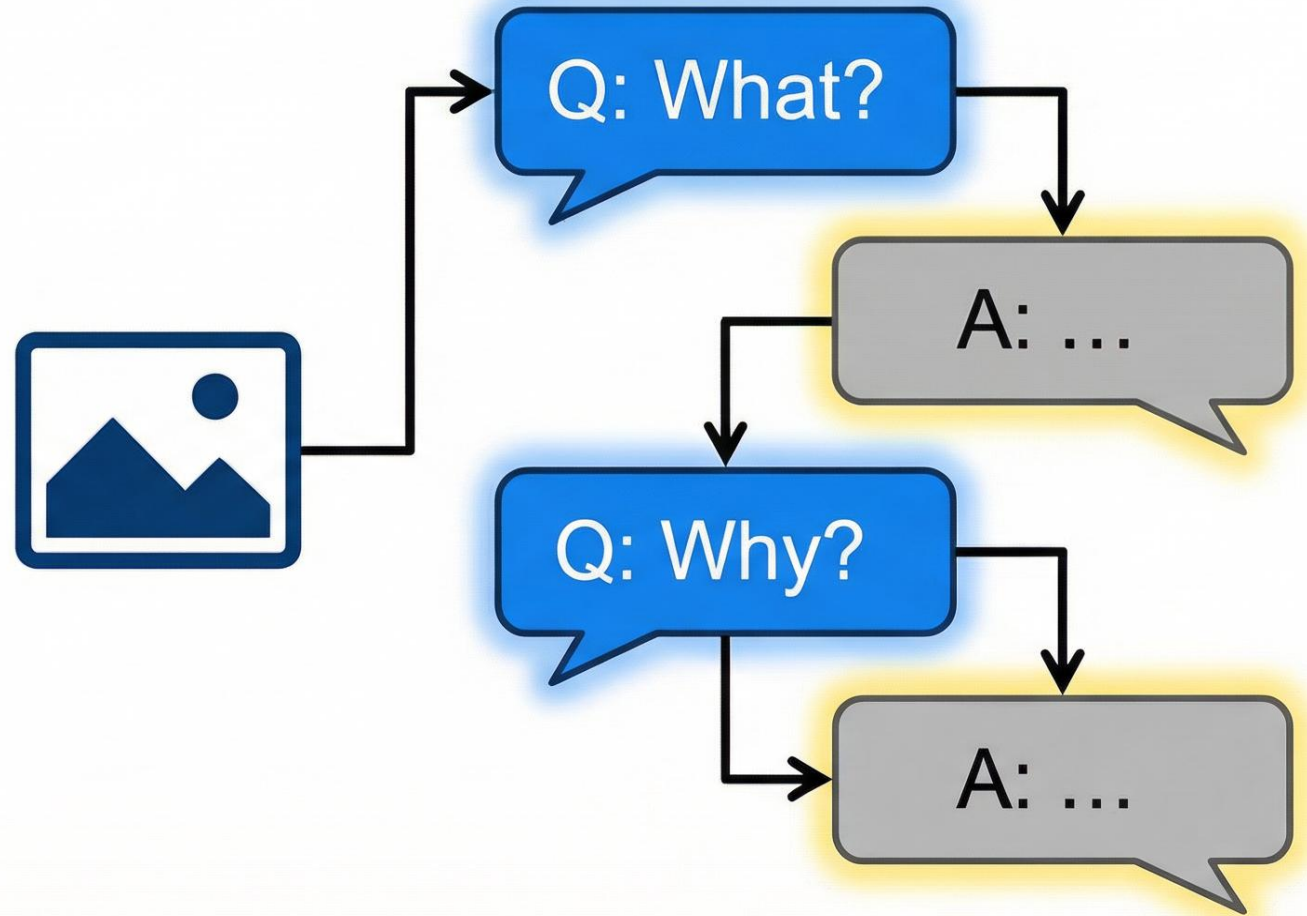
Trend 2: Zero-shot



Example: CLIP (OpenAI)

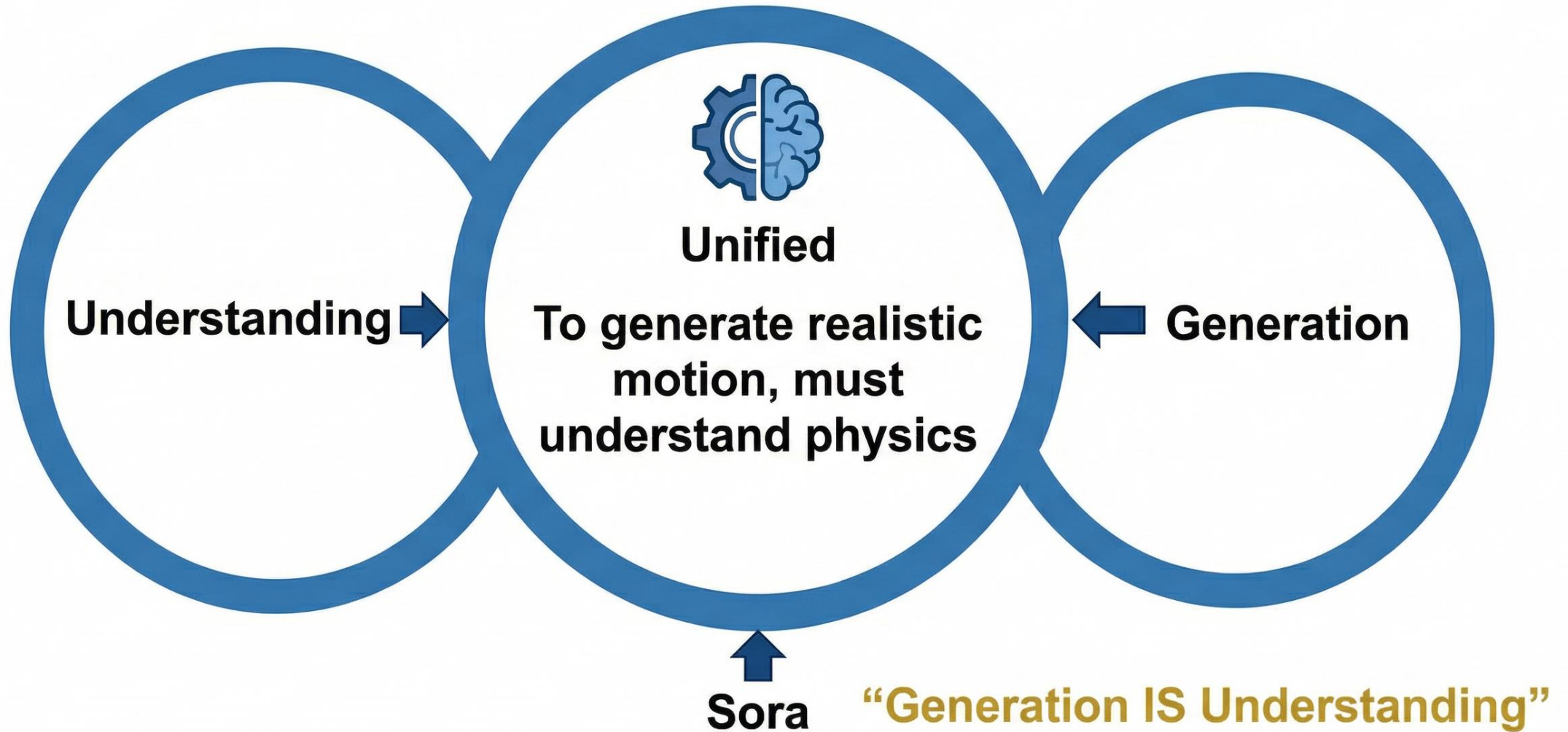
Trend 3: Visual Dialog

- Answer any question
- Multi-turn reasoning
- From recognition → understanding

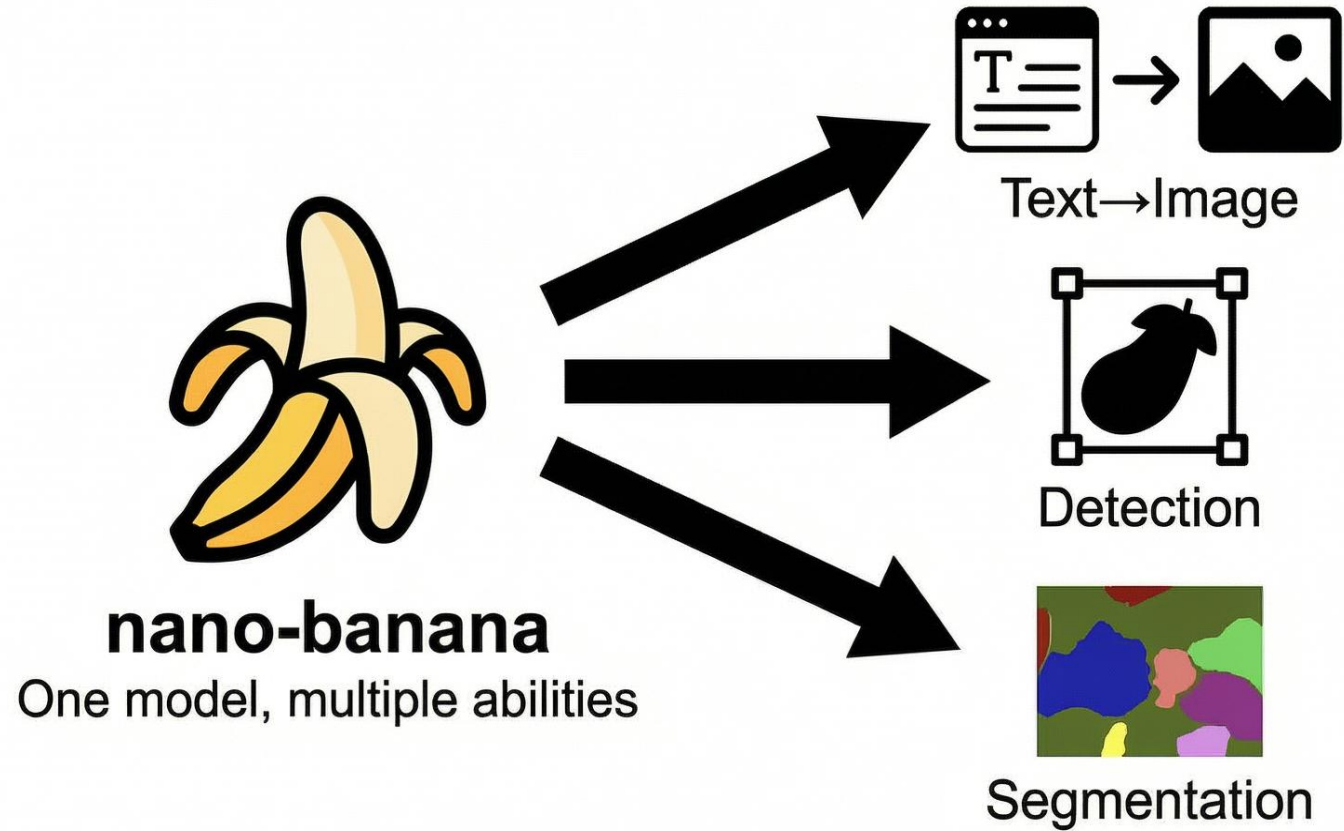


GPT-4V, Gemini, Claude

Trend 4: World Model



Live Demo: nano-banana



Let's see it!





detect all faces with red boxes

◆ 显示思路 (Nano Banana Pro) ▼



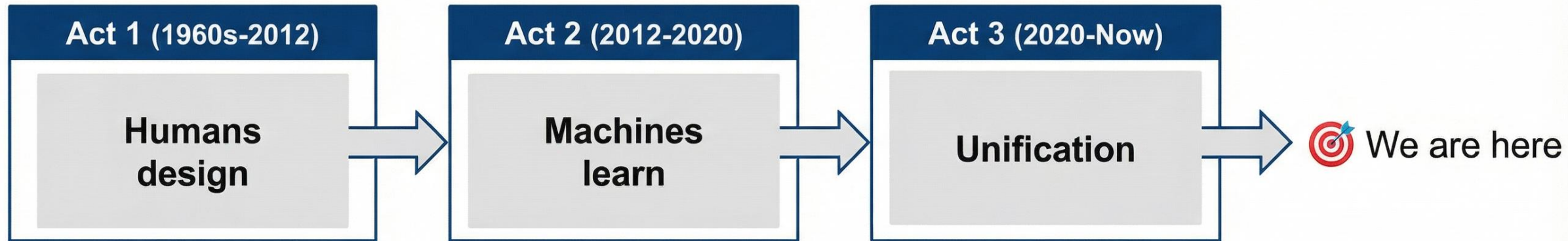
Please use different colors to segment all person in the image.

显示思路 (Nano Banana Pro) ▼





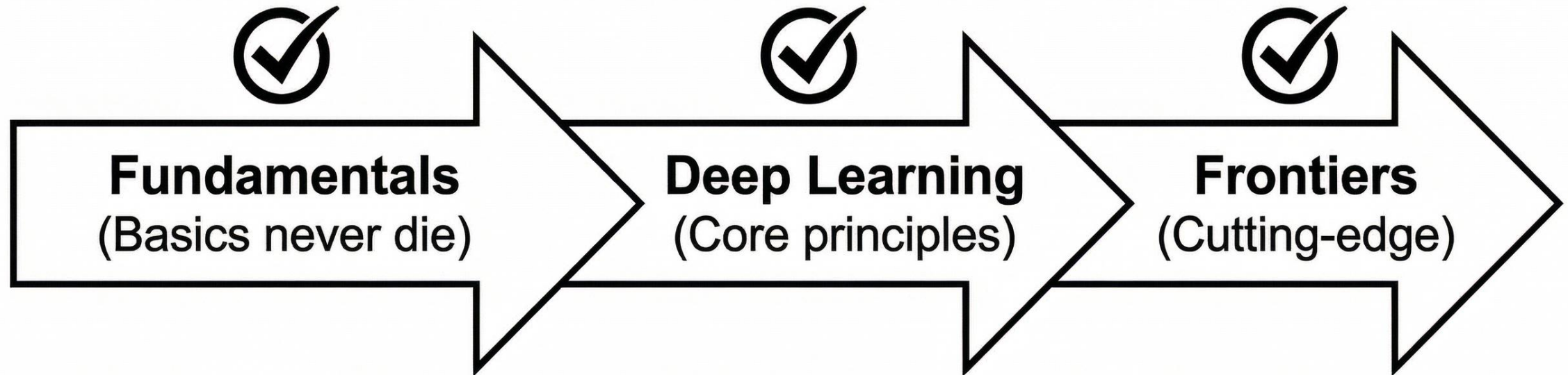
Three Eras: Complete Timeline



Each builds on last



Why Fundamentals?



- ✓ Basics never die
- ✓ Understand principles
- ✓ DL builds on them

Weeks 1-4 → 5-9 → 10-12

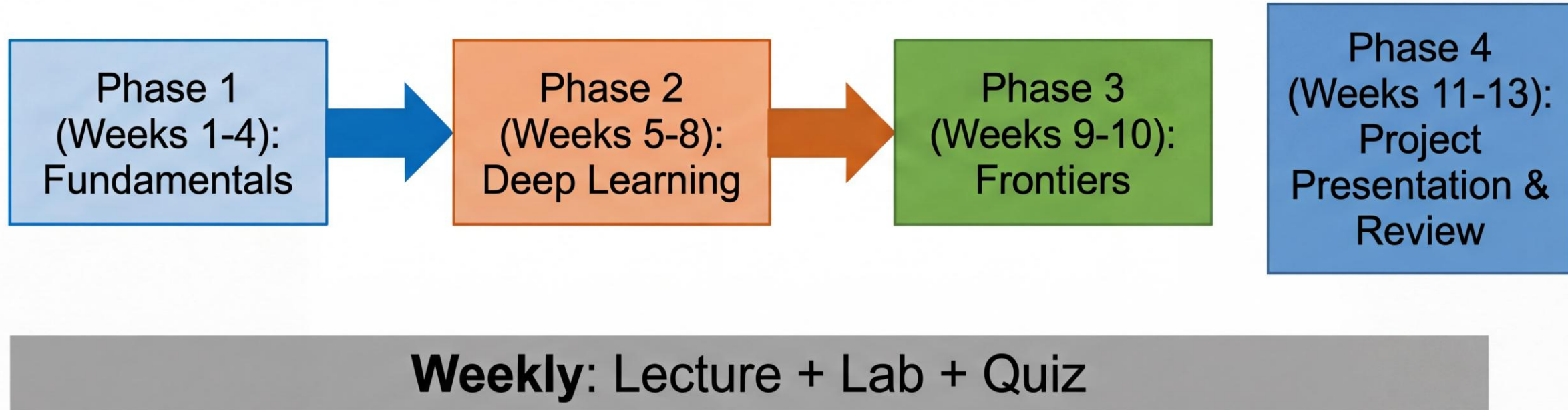
Course: Fundamentals → DL → Frontiers

Part 3

Course Overview



课程：13周



Phase 1: Fundamentals

Week 1: Intro & Image Basics

Week 2: Image Filtering

Week 3: Edge Detection

Week 4: Feature Extraction

Goal: Build foundation



Phase 2: Deep Learning

Week 5: Machine Learning Basics for Vision

Week 6: Neural Networks (MLP) & CNN Basics

Week 7: Training Strategies & Data

Goal: Master deep learning based CV methods.



Phase 3: Frontiers

Week 8: Object Detection & Segmentation

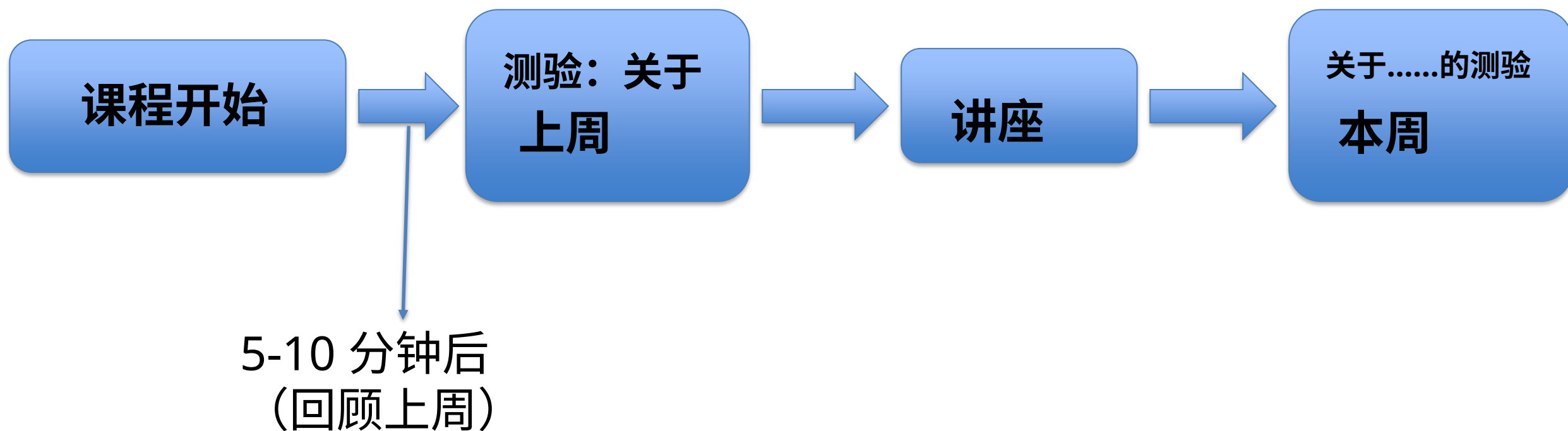
Week 9: Vision Transformers (ViT)

Week 10: Generative AI & Frontiers



Goal: Explore cutting-edge CV methods.

每周 结构



前 5-10 分钟预留出来，以便迟到者签到。

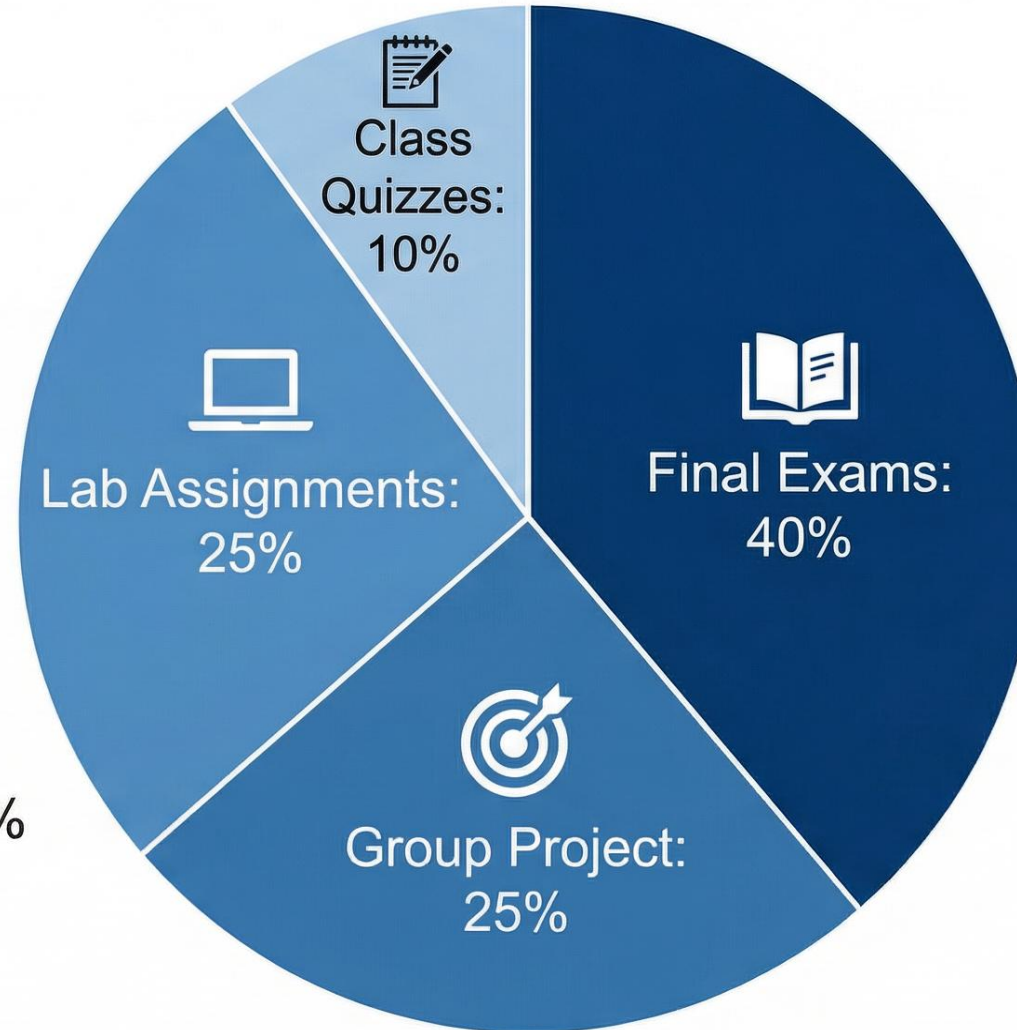


Part 4

Grading & Assessment



Grading Breakdown



-  Class Quizzes: 10%
-  Lab Assignments: 25%
-  Group Project: 25%
-  Final Exams: 40%

Total: 100%



Class Quizzes: 10%

Frequency:	Almost weekly
Format:	MC, 5-10min
Scoring:	Average all (0 for absence)



Purpose: Maintain
pace, identify gaps
(not penalize)



Lab Assignments: 25%

Weekly (~12 labs)
Pass/Fail (Complete = Pass)
Scoring: # Pass / Total

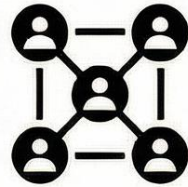
No pressure! Just complete.



Group Project: 25%



Team: 2 students



Week 3-4:
Team
Formation



Week 5-6:
Topic
Selection



W12 & W13:
Presentation



Content: CV
task
implementation

Grading Dimensions

- Implementation
- Innovation
- Presentation



Assessment Breakdown



Class Quizzes: 10%



Lab Assignments: 25%



Group Project: 25%



Final Exams: **40%**

Total: 100%

Balanced: 60% continuous + 40% final



Academic Integrity



Lab

Discuss

OK, own code



Team

OK, no copying



Quiz/Exam

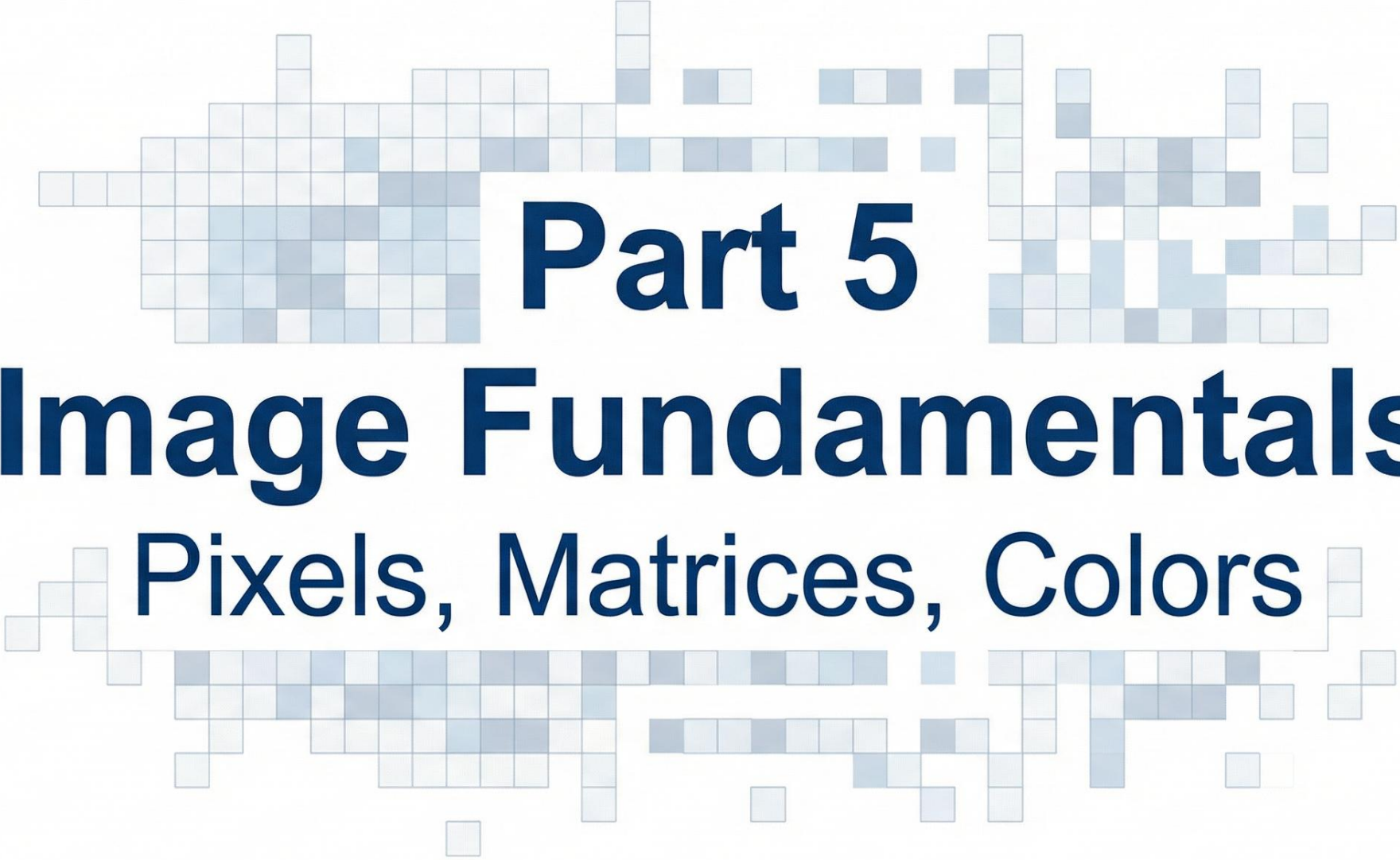
Individual work



Violation: Serious consequences

Be honest, be fair.





Part 5

Image Fundamentals

Pixels, Matrices, Colors

What Computer Sees?



Gap

[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[16, 61, 32]	[46, 61, 69]	[46, 61, 39]	[123, 45, 67]	[123, 45, 100]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[16, 49, 35]	[46, 61, 66]	[49, 49, 40]	[123, 45, 67]	[123, 45, 100]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[18, 63, 31]	[58, 61, 63]	[48, 63, 66]	[123, 45, 67]	[123, 45, 100]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[26, 53, 33]	[59, 43, 33]	[48, 63, 69]	[123, 45, 67]	[123, 45, 100]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[89, 167, 22]	[67, 41, 31]	[49, 69, 48]	[123, 45, 67]	[123, 45, 103]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[169, 167, 22]	[188, 163, ...]	[80, 160, 22]	[108, 168, ...]	[123, 45, 100]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22]	[46, 67, 31]	[39, 64, 29]	[82, 64, 22]	[122, 46, 42]	[123, 45, 100]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22]	[40, 93, 21]	[90, 92, 21]	[32, 63, 24]	[37, 72, 18]	[123, 45, 106]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22]	[90, 91, 21]	[89, 168, 22]	[59, 93, 64]	[123, 46, 42]	[123, 46, 108]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22],	[89, 160, 21]	[59, 160, 21]	[59, 99, 24]	[123, 46, 57]	[122, 46, 108]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22]	[89, 169, 22]	[59, 163, 22]	[39, 59, 21]	[122, 46, 55]	[122, 46, 991]
[123, 45, 67],	[200, 10, 59],	[89, 167, 22],	[89, 167, 22]	[89, 169, 27]	[89, 169, 28]	[79, 92, 35]	[123, 46, 67]	[123, 46, 201]
[123, 45, 67],	[200, 10, 59],	[89, 167, 22],	[89, 167, 22]	[99, 167, 29]	[80, 99, 26]	[76, 52, 31]	[123, 46, 61]	[123, 46, 201]
[123, 45, 67],	[200, 10, 59],	[89, 167, 22],	[89, 167, 22]	[89, 183, 33]	[80, 189, 23]	[79, 72, 30]	[123, 46, 62]	[123, 46, 271]
[123, 45, 67],	[200, 10, 50],	[89, 167, 22],	[89, 167, 22]	[89, 99, 29]	[80, 183, 31]	[80, 97, 33]	[120, 45, 67]	[123, 45, 221]

You: Beach, Waves, Sunshine, Happiness

Computer: [123, 45, 67]...

This gap = where CV begins



Image = Matrix



=

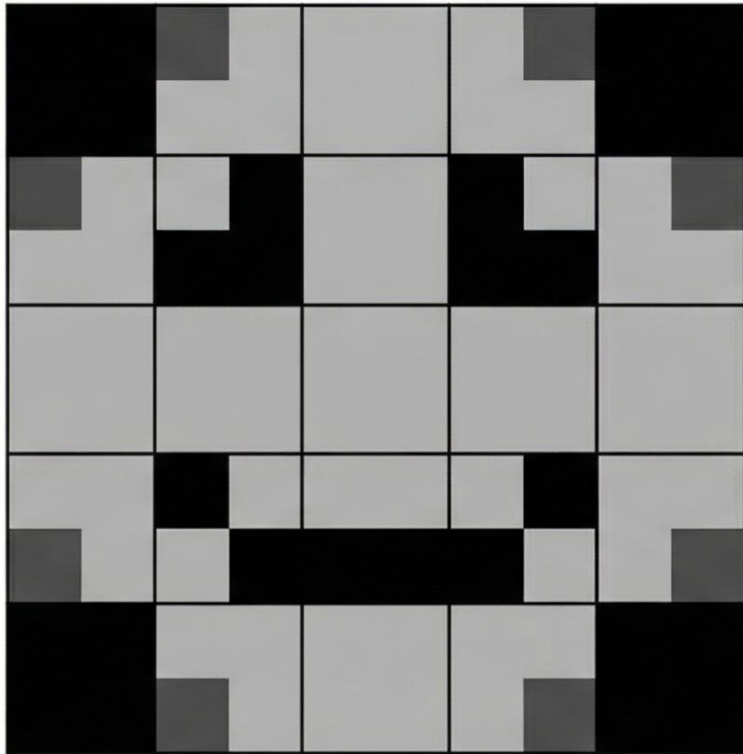
255	200	255
200	50	200
255	200	255

Each pixel = number
(grayscale) or 3
numbers (color)

Range: 0-255

Everything is
numbers.

Grayscale Example



Single channel:
1 number per
pixel.

0 = Black.
255 = White.

5x5 matrix.

0	255	255	0	0
0	255	113	255	0
0	255	113	255	0
0	255	113	255	0
0	0	0	0	0

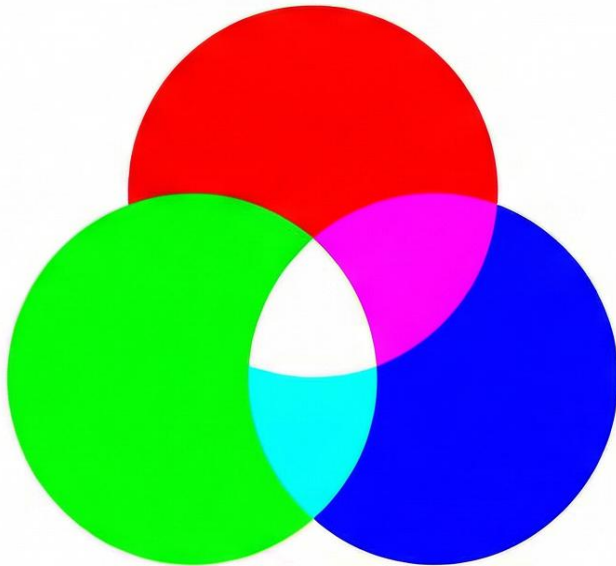
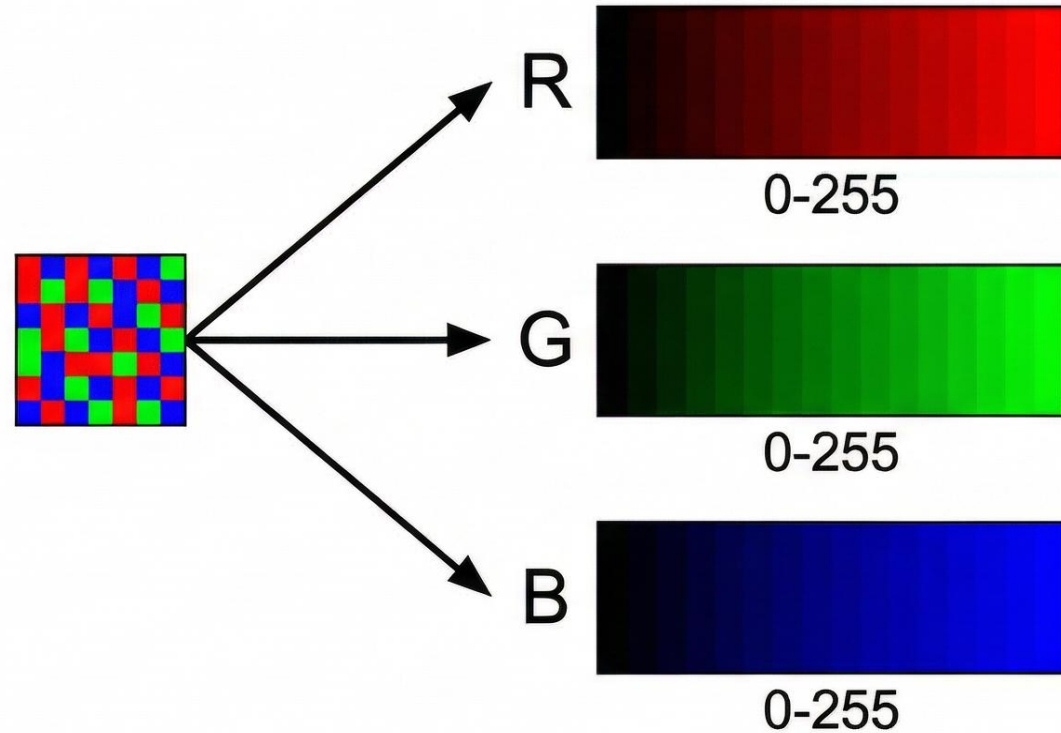
0 (Black)



→ 255 (White)

Color = 3 Channels (RGB)

Color = 3 Channels
Each pixel = (R, G, B)
Each: 0-255

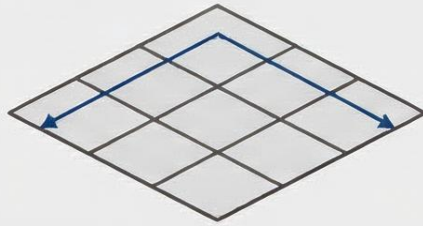


→ Red + Green + Blue → All colors

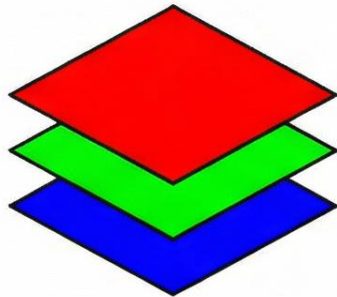
Dimensions & Resolution

Dimensions

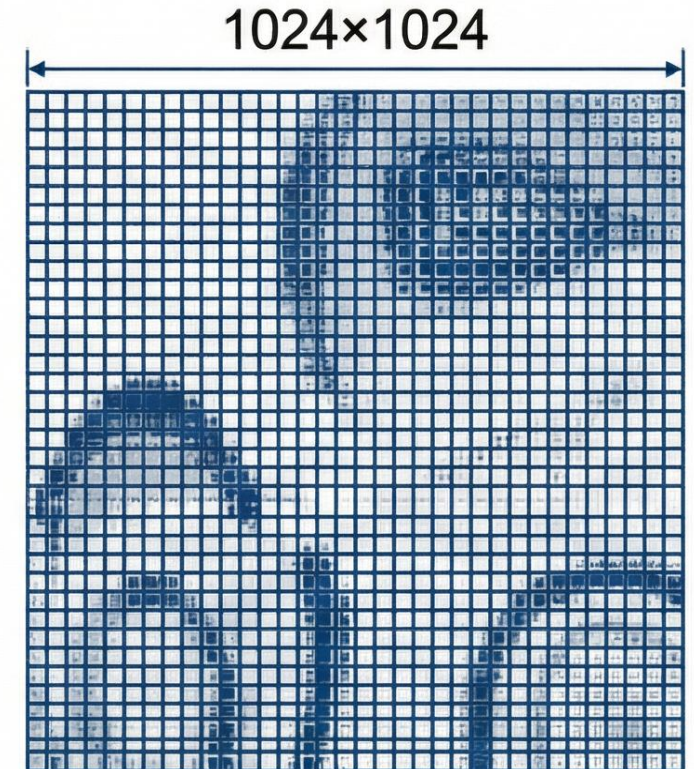
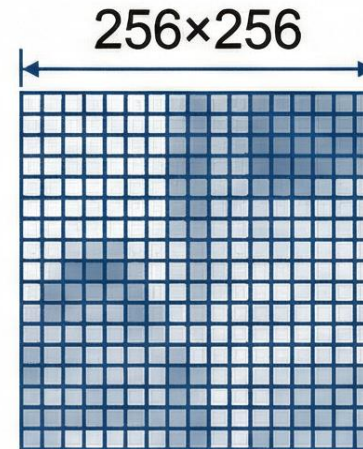
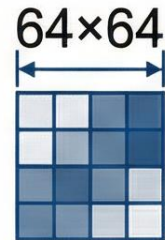
Grayscale: $H \times W \times 1$



Color:
 $H \times W \times 3$



Resolution: Total pixels



Higher resolution → More details, Larger file

Demo Code: Load & Inspect

```
import cv2

img = cv2.imread('photo.jpg')
print(img.shape)
# (1080, 1920, 3)

pixel = img[100, 200]
print(pixel)
# [B, G, R]
```

OpenCV uses BGR order



Demo Code: Split Channels

```
b, g, r = cv2.split(img)
print(b.shape)
# (1080, 1920)
```

```
cv2.imshow("R", r)
cv2.imshow("G", g)
cv2.imshow("B", b) '
```

R



G

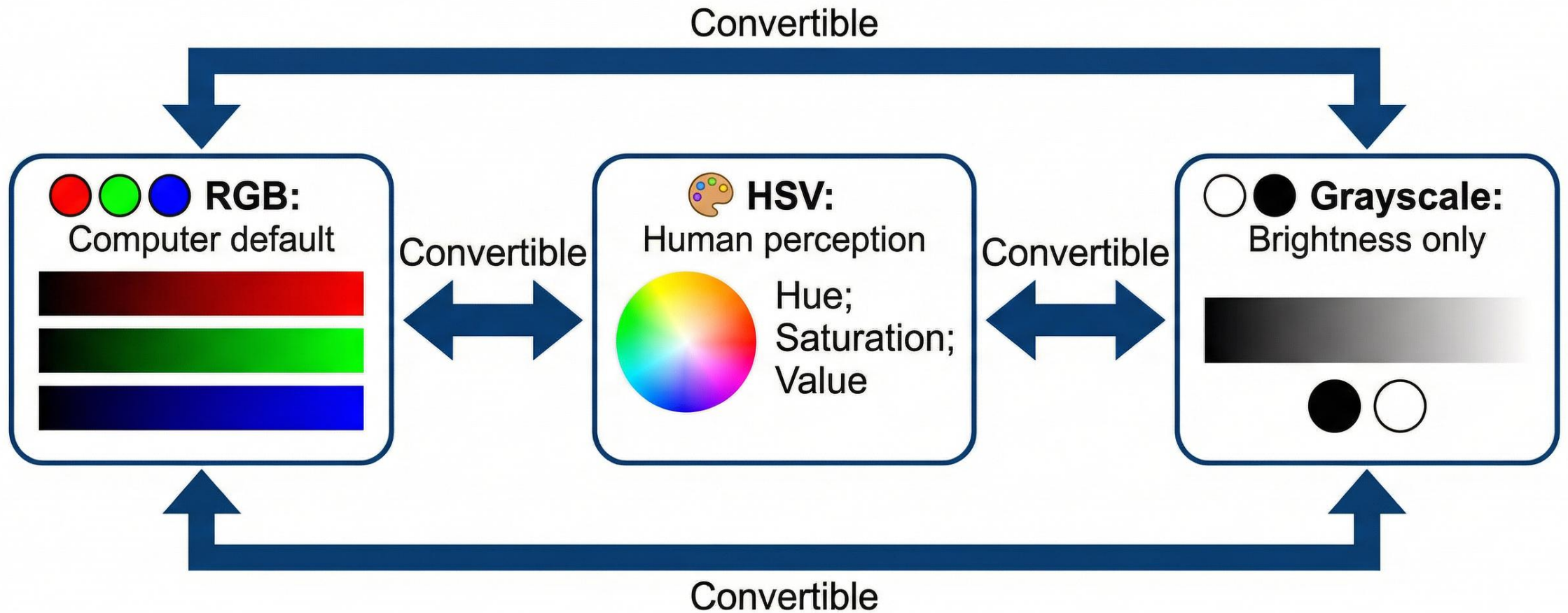


B



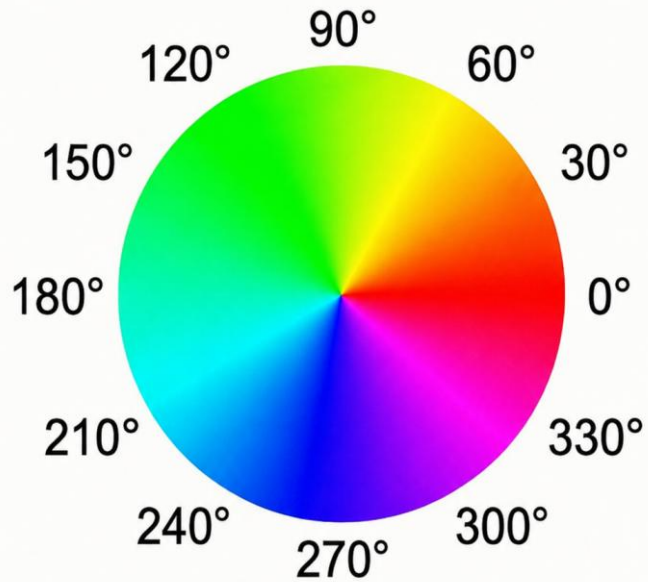
Each = grayscale.

Color Spaces



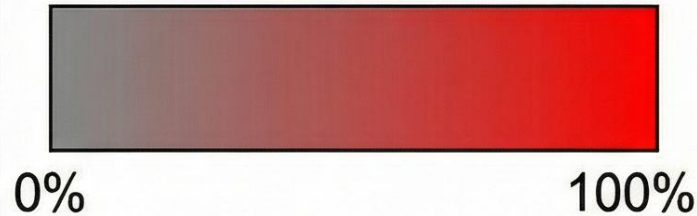
HSV Color Format

HUE (H): Color Type



0-360°: Defines base color
(e.g., Red, Blue, Green).

SATURATION (S): Color Purity



0-100%: Intensity.
Low=Gray, High=Vibrant.

0-100%: Intensity.
Low=Gray, High=Vibrant.

VALUE (V): Brightness



0-100%: Lightness.
Low=Black, High=White.

0-100%: Lightness.
Low=Black, High=White.



Color Space Conversion

RGB -> HSV

逻辑：将 RGB 立方体投影到圆柱体（V=最大值，S=范围，H=角度）

V, S, H = max(R,G,B), (max(R,G,B)-min(R,G,B))/max(R,G,B), get_angle(R,G,B)_

hsv = merge(H, S, V)

RGB -> 灰度

逻辑：基于人眼敏感度的加权求和（绿色>红色>蓝色）

灰色 = $0.114 * \text{蓝色} + 0.587 * \text{绿色} + 0.299 * \text{红色}$

print(gray.shape)# (H, W) 灰度图只
有二维



Color Space Conversion

```
# RGB → HSV
hsv = cv2.cvtColor(img,
cv2.COLOR_BGR2HSV)

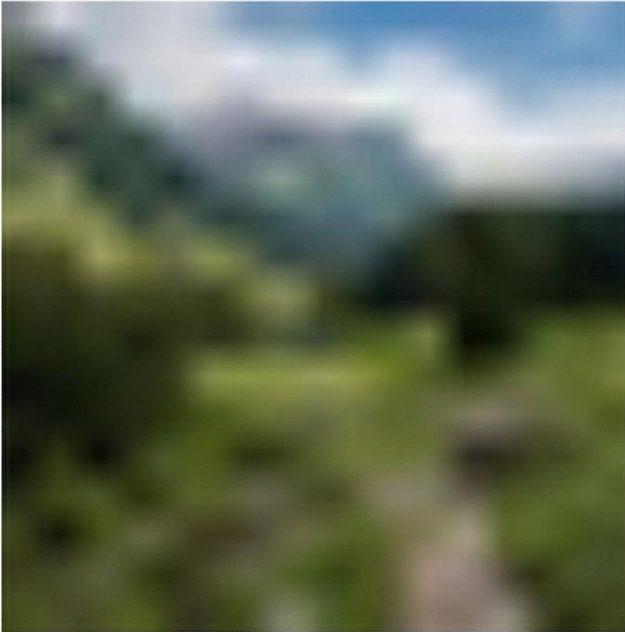
# RGB → Gray
gray = cv2.cvtColor(img,
cv2.COLOR_BGR2GRAY)

print(gray.shape)
# (H, W)
Grayscale has 2D only
```



Resolution Matters?

Resolution Impact



Low (32x32): Fewer details,
Smaller, Faster



High (1024x1024): More
details, Larger, Slower

Trade-off: Task needs vs Resources



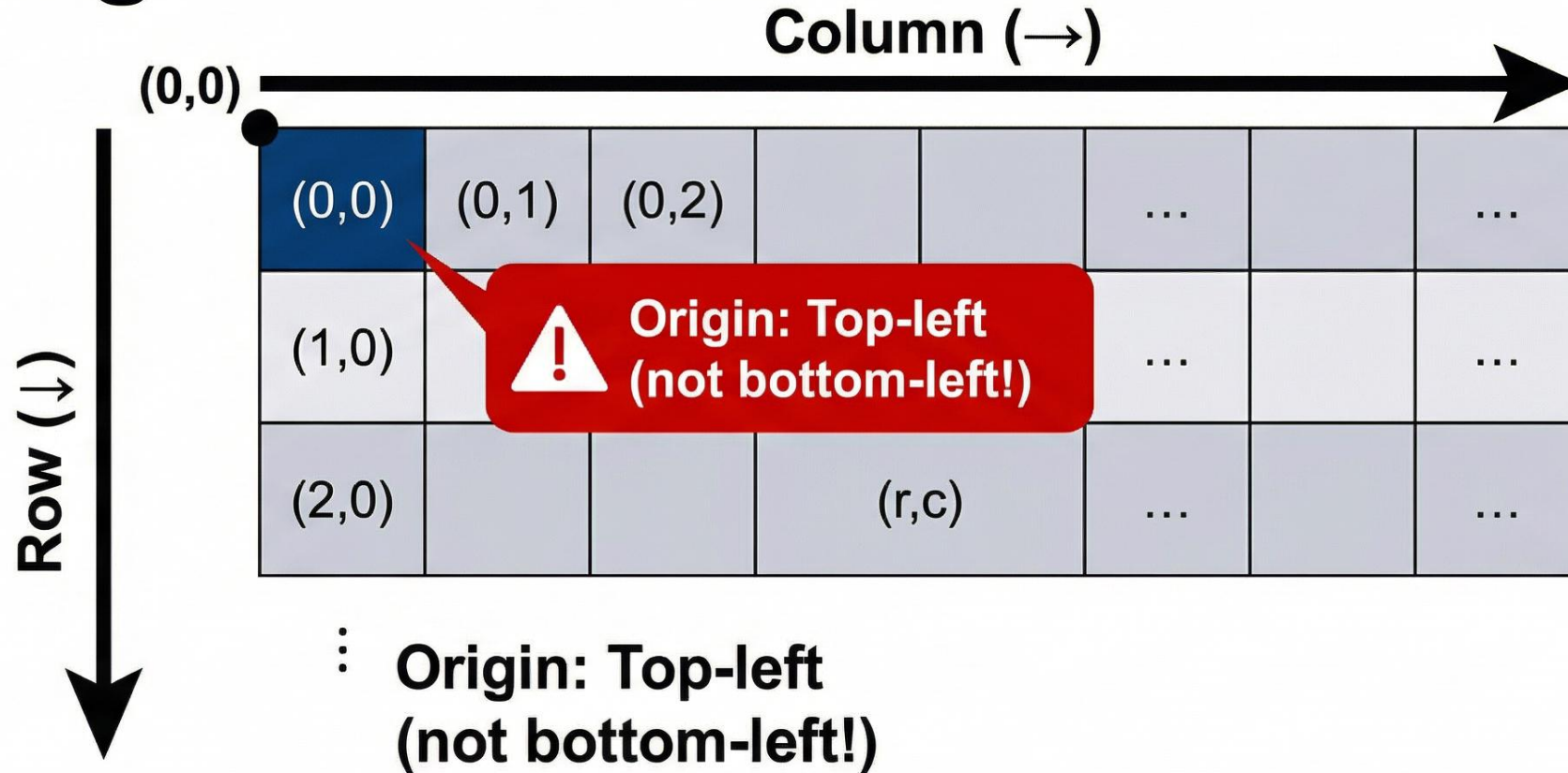
Image Fundamentals: Summary

- ✓ Image = Matrix of numbers
- ✓ Grayscale: $H \times W \times 1$ (0-255)
- ✓ Color: $H \times W \times 3$ (RGB)
- ✓ Color spaces convertible
- ✓ Resolution = pixel count

“Everything is numbers”



Image Coordinates



First index: Row (↓)
Second index: Col (→)

**img[row, col]
= img[y, x]**

Common Mistake

✗ Wrong:

`img[x, y]`



Correct

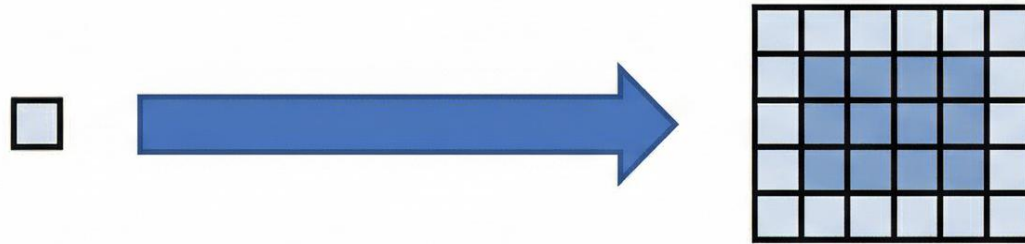
✓ Correct:

`img[y, x]` ✓
`img[row, col]`

Memory: `y` = vertical (row), `x` = horizontal (col)

Next Week: Image Filtering

Concept: Pixel-wise $\rightarrow$ Neighborhood



Topics: Blur, Sharpen, Denoise, Convolution



Lab: Implement filters



See you!

有用信息

- 顶级会议
 - 计算机视觉与模式识别 (CVPR) – 国际计算机视觉会议 (ICCV) – 欧洲计算机视觉会议 (ECCV) – 国际学习表征会议 (ICLR) – 神经信息处理系统 (NeurIPS) – 国际机器学习会议 (ICML)
- 顶级期刊
 - IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) – International Journal of Computer Vision



实用信息

- Google Scholar –
论文 · 相关工作

- 引用 – 作者 –

Gemini Deep Research

- Arxiv (这些论文未经同行评审!)
- 学者/项目主页
- Github – 代码 (不要全信! 用简单示例测试其可用性) – 特定领域的论文合集 (搜索“awesome-**”)
- Youtube – 顶级会议的口头报告 – 教程



Closing Quiz:

Purpose: Attendance + Key info check

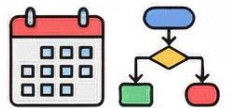
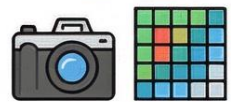
Easy questions, don't worry!



Week 1: Complete! 🎉



匿名 课堂反馈

- ✓ Applications 
- ✓ History 
- ✓ Course Plan 
- ✓ Image Basics 

Today's Lab:

Setup + Instagram Filters 

Next: Image Filtering 

See you next week!

